

MAJOR REVIEW

Computer Vision Syndrome: A Review

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Abstract. As computers become part of our everyday life, more and more people are experiencing a variety of ocular symptoms related to computer use. These include eyestrain, tired eyes, irritation, redness, blurred vision, and double vision, collectively referred to as computer vision syndrome. This article describes both the characteristics and treatment modalities that are available at this time. Computer vision syndrome symptoms may be the cause of ocular (ocular-surface abnormalities or accommodative spasms) and/or extraocular (ergonomic) etiologies. However, the major contributor to computer vision syndrome symptoms by far appears to be dry eye. The visual effects of various display characteristics such as lighting, glare, display quality, refresh rates, and radiation are also discussed. Treatment requires a multidirectional approach combining ocular therapy with adjustment of the workstation. Proper lighting, anti-glare filters, ergonomic positioning of computer monitor and regular work breaks may help improve visual comfort. Lubricating eye drops and special computer glasses help relieve ocular surface-related symptoms. More work needs to be done to specifically define the processes that cause computer vision syndrome and to develop and improve effective treatments that successfully address these causes. (*Surv Ophthalmol* 50:253–262, 2005. © 2005 Elsevier Inc. All rights reserved.)

Key words. Asthenopia • computer vision syndrome • dry eye • ergonomics • eyestrain • glare • video display terminals

I. Introduction

Twenty years ago, the advent of computers revolutionized the workplace. Until that time, office work had involved a range of activities including typing, filing, reading, and writing. Each activity was adequately varied in the requirements of posture and vision, posing a natural “break” from the previous activity. The introduction of computers, however, has combined these tasks to where most can be performed without moving from the desktop, thereby improving quality, production, and efficiency. In fact, it is estimated that the 75% of all jobs in the year

2000 involved computer usage.¹⁶ The popularity and affordability of personal computers with Internet capabilities at home has introduced even more computer users. In 1990, about 15% of U.S. households owned a computer, a number that has now increased to 50% of all households.⁷³

Because of this extensive use of computers many studies have been conducted in an attempt to address questions concerning safety and health for video display terminal (VDT) users. The large majority of research has addressed the question of radiation levels from VDTs, known to emit many types of radiation

including x-radiation, optical radiation, radio frequency radiation, very low frequency radiation, and extremely low frequency radiation.^{74,93} Studies have not clearly indicated a negative effect on the computer user (Abelson MB: How to fight Computer Vision Syndrome. Rev Ophthalmol 114–6, 1999). During the late 1980s and early 1990s, concern of possible reproductive effects from using VDTs arose following reports of adverse pregnancy outcomes among groups of women computer users. A recent review concluded that for most women in modern offices, work with VDTs does not increase their risk of miscarriage.^{3,21,30,44,55} Another study has even revealed that somatic disorders, depression, and obsessions are increased with computer usage, especially when operating time is more than 30 hours per week and usage of more than 10 years.⁹⁵

Studies have shown, however, that eye-related symptoms are the most frequently occurring health problems among VDT users.^{15,16,18,80,83,87} The main visual symptoms reported by VDT users include eyestrain, tired eyes, irritation, burning sensation, redness, blurred vision, and double vision,^{4,5,13,15,18,52,85,87} thus termed the phrase “Computer Vision Syndrome” (CVS). In 1992, a total of 1,307 surveys were completed by optometrists who reported that the majority of VDT patients have symptoms that are different than other near-point workers, especially as related to glare, lighting, unique viewing conditions, and spectacle requirements. Greater frequency and severity of symptoms were also noted.⁸⁰ Travers and Stanton identified a trend in symptomatology, whereby symptoms appeared to increase as duration of VDT exposure increased.⁸⁸ It is estimated according to some reports that the diagnosis and treatment of these symptoms costs almost US\$ 2 billion each year (Abelson MB: How to fight Computer Vision Syndrome. Rev Ophthalmol 114–6, 1999). As computer users become more aware of CVS, it is important that ophthalmologists are attentive to this rapidly evolving disorder, as we could be facing a possible epidemic of the 21st century.

The purpose of this review is to examine the epidemiology, causes, and diagnosis of CVS; we will also present a review of the current treatments for CVS.

II. Definition

The ocular complaints experienced by computer users typically include eyestrain, eye fatigue, burning sensations, irritation, redness, blurred vision, and dry eyes, among others. The condition of a person experiencing one or more of these ocular complaints as a result of operating a computer and looking at a computer monitor is generally referred to as “Computer Vision Syndrome” (CVS). It is a repetitive strain

disorder that appears to be growing rapidly, with some studies estimating that 90% of the 70 million U.S. workers using computers for more than 3 hours per day experience it in some form. Non-ocular symptoms include headaches, pain in the shoulders, neck, or back. As diverse as the symptoms are, they may be related and can be subdivided into three potential pathophysiological causes: 1) ocular surface mechanisms, 2) accommodative mechanisms, and 3) extraocular mechanisms. There is a significant gap in the fund of knowledge regarding these pathophysiological factors that cause this disorder.

III. Symptomatology

It is quite clear from several studies that use of a VDT causes asthenopia. In fact, visual complaints were reported by 75% of VDT operators working 6–9 hours in front of their screens compared to 50% of other workers.⁶⁰ Surveys of optometrists in the U.S. and the United Kingdom indicate that 12.4% and 9.0% of their patients, respectively, are examined primarily because of symptomatic visual or ocular problems associated with using a computer.⁶⁰ Table 1 shows a categorization of the most common symptoms which are eye-related.

A. ASTHENOPIC SYMPTOMS

Many individuals have marginal vision disorders, such as difficulties with accommodation or binocular vision problems that do not cause symptoms when performing less demanding visual tasks.⁷⁹ However,

TABLE 1
*Computer-related Vision Symptoms
and Common Diagnoses*

Symptom Category	Symptoms	Diagnosis
Asthenopic	Eyestrain Tired eyes Sore eyes	Binocular vision Accommodation
Ocular surface-related	Dry eyes Watery eyes Irritated eyes Contact lens problems	
Visual	Blurred vision Slowness of focus change Double vision Presbyopia	Refractive error Accommodation Binocular vision
Extraocular	Neck pain Back pain Shoulder pain	Presbyopic correction Computer screen location

prolonged VDT usage has been shown to cause diminished power of accommodation, removal of the near point of convergence, and deviation of phoria for near vision.⁸⁹ The results of this suggested that weakness of these important visual functions could be the cause of eyestrain in computer operators.⁸⁹ In another study subjects over-accommodated by an average of -0.50 to -0.75 D when stimuli were placed at 40 cm, and by -0.75 D to colored letters on a colored background.^{8,66} Prolonged work at a VDT has been reported to result in changes in both relative accommodation and vergence.^{7,17,19,31,33}

Gur and Ron evaluated the prevalence of vision problems in VDT workers as well as the effect of 4 days of VDT use on near-point accommodation (NPA). NPA for VDT users and non-users was measured at the beginning of the day on the start of the workweek, and again at the end of the day 4 days later. Interestingly, a high prevalence of exophoria, convergence insufficiency, and low fusional convergence were found among VDT workers. They also found that the accommodative amplitude decreased significantly for VDT users (by 0.69 D) than non-users (0.18 D) between the first examination and the second examination 4 days later (Table 2).³³

In another longitudinal study Yeow and Taylor reported that subjects below age 40 years who used VDTs lost more accommodative amplitude than who did not.^{100,101}

In summary, changes in accommodative and vergence functions have been reported to occur after work periods at a VDT, and these changes have been proposed as objective indicators of subjective visual fatigue. They are also most likely transient, with workers returning to baseline values by the end of the workday or week. Substantial losses in these functions would otherwise be expected in long-term studies. Such studies have not found a difference at least in NPA and near-point convergence between VDT users and nonusers. VDTs are a major source of near work for many adults but do not appear to result in losses

of accommodative and vergence function beyond the ordinary effects of age.

1. VDTs and Transient Myopia

Accommodative effort during near work is thought to be a causative factor in the development of myopia.⁶⁰ Although it is questionable whether VDTs are associated with a risk of myopic progression in adults compared to paperwork, it is clear that near work with VDTs results in a small, temporary myopic shift. In a cross-sectional comparison of VDT users and typists, VDT users experienced a myopic shift of about -0.12 D after the work period, while the refractive error of typist was unchanged.^{69,70,100} The shifts were too small to affect distant visual acuity. Luberto et al observed a transient myopia in 20% of VDT workers at the end of their work shift.⁵⁴ All subjects undergoing this myopic change complained of asthenopia, but only 32.5% of those with asthenopia experienced the transient myopic shift.⁵⁴

There is no compelling evidence in the literature that states there is a significant increase in the risk of myopia onset or progression from the use of VDTs in adults compared to other forms of near work. A small, transient myopic shift appears to occur after VDT use, but its significance with respect to creating permanent myopic change is unknown.⁶⁰

B. OCULAR SURFACE RELATED SYMPTOMS

Computer users often report complaints of eye dryness, burning, grittiness, or heaviness after an extended period of time at the terminal. Users' eyes may even tear in an attempt to restore the proper chemical balance and to properly lubricate and rewet the front surface of the eye. Dry eye may be a primary cause of ocular fatigue, such as experienced when using a VDT when the blink rate is decreased and the exposed ocular surface area is increased, causing desiccation of the eye. It has been postulated that the blink rate is decreased further in a dark setting where it is difficult to read, and that accelerated desiccation may be responsible for the fatigue.¹ Several other factors contribute to drying of the ocular surface, including the following categories.

1. Environmental Factors

The cornea is very sensitive to drying and chemical imbalances from environmental factors. The office includes hazards such as dry air, ventilation fans, static buildup, airborne paper dust, laser and photocopy toner, and building contaminants.^{11,84,91}

2. Reduced Blink Rate

Most individuals normally blink between 10–15 times per minute. Studies have shown that the blink

TABLE 2
*Prevalence of Visual Impairments in VDT
Users And Nonusers*

Type of Impairment	VDT % (n = 32)	Control % (n = 15)
Low fusional convergence	46.9	13.3
Heterophoria	34.4	13.3
Convergence insufficiency	28.1	13.3
Amblyopic eye	12.5	0
Refractive disorders	12.5	0
Suspected eye disease	9.4	6.6
Color blindness	6.2	6.6
No stereopsis	6.2	6.6

Based on Gur et al.³³

rate at the computer is significantly less than normal.^{1,64,99} A reduced blink rate at the VDT contributes to a poor tear film quality and temporary stresses the cornea, resulting in symptoms of dry eye. This reduction in blink rate may as great as 60% and be also be one of the basic pathophysiologic mechanism for the high incidence of meibomian gland disease in computer users with significant ocular surface complaints (authors' data).

3. Increased Exposure

The reading of text on paper is normally performed while looking downwards. This results in the eyelid covering a substantial portion of the front surface of the eye, thus minimizing the evaporation of tears. On the contrary, computer users usually view their reading material in a horizontal gaze. This results in a wider palpebral fissure and an increased surface area exposed to the effects of evaporation.

4. Sex

The prevalence of dry eye is slightly greater in females than males.^{71,81}

5. Age

Tear production normally decreases with age. Although dry eye can occur at any age in both men and women, post-menopausal women represent the group of individuals most affected by dry eye.^{71,81}

6. Systemic Diseases and Disease Syndromes Associated with Dry Eye

Dry eye is associated with various systemic diseases. A general review of the patient's entire medical history aids in establishing a complete diagnosis. In the case of Sjogren syndrome, dry mouth and rheumatoid arthritis can assist in making the diagnosis. Several autoimmune diseases also have association with dry eye.⁵³

7. Systemic Medications

There are several systemic medications contributing to ocular drying. The most prevalent medications include diuretics, anti-histamines, psychotropics, and anti-hypertensives.⁶⁷

8. Contact Lens Use

Office workers wearing contact lenses were found to be more likely to suffer a higher severity of ocular discomfort.^{81,96} Contact lens comfort is highly dependent on lubrication of the eye. The contact lens surface should skate along the eye and eyelid surfaces with minimal resistance. If the ocular surface is dry, the lenses dry and adhere to the upper eyelid during

the blink. This "friction effect" from dry eye produces the discomfort described.^{81,96}

9. Ocular Conditions

Localized dysfunctions of the glands, which produce the ocular tear film, can also contribute to dry eye. The most common disorder, anterior blepharitis, is an inflammation of the eyelids, affecting the meibomian glands that secrete the lipid layer of the ocular surface. The lack of an adequate lipid layer contributes to rapid evaporation of the water component of the tear film thus causing discomfort.⁵³

10. Cosmetics

Poorly applied cosmetics can block the openings of the oil-secreting meibomian glands. This in turn contributes to a rapid evaporation of the water component of the tear film and the resultant discomfort (www.brower.co.uk/opticians/dryeyes.html).

IV. Visual Effects of Display Characteristics

A. DISPLAY QUALITY

The National Research Council Committee on Vision stated that "poor display quality ... probably contributes to the annoyance and discomfort sometimes reported by workers. ... Visual performance is affected by a number of display parameters, such as character size, structure, and style; and by image contrast and stability."⁶⁷ There appears to be little disagreement regarding the effect of monitor design and display quality on visual performance.

The images that are produced on a video display terminal consist of thousands of tiny, bright spots (pixels) or horizontal lines (rasters) that collectively form unresolved images that blur together and lack sharp edges. The more dots or lines displayed on a monitor to produce a picture, the sharper and clearer the image will appear. It is thought that slightly blurred characters would create an understimulation of accommodation, creating a lag of accommodation behind the image on the screen.⁴⁰ Ziefle also noted a functional characteristic of computer operators and resolution through the comparison of monitor resolutions at 62 dots per inch (dpi) and 89 dpi. She determined that search reaction times and fixation durations when viewing documents were significantly increased with the lower resolution.¹⁰² In addition, the extent of visual fatigue correlated with both search reaction times and eye movement parameters. Fortunately, over the past decade the resolution of monitors has improved drastically, producing displays approaching that of typeset documents.⁸⁷

Several factors affect readability and legibility of characters displayed on the screen. Words containing upper case in combination with lower case are more easily interpreted than all upper-case documents.³⁵ The spacing between characters and lines also affects picture quality and should allow for at least one-half character space between words and one character space between lines.¹⁶ High levels of contrast and brightness are known to represent the most common causes of character blur. It is also recommended that screens contain dark characters against a light background display screen rather than the opposite.^{57,58,73,86,87} When a VDT operator constantly switches from a light background hard copy to dark background display, fatigue of the iris muscle can result.^{10,86}

B. LIGHTING AND GLARE

Improper lighting conditions of a workstation can also adversely affect a VDT user's ocular comfort.^{6,22,26,27,34,36,39,49,68,77,78,86,97} Constant and bright illumination from surrounding sources of light (overhead fluorescent, large windows, desk lamps) appears to wash out screen character images, creating reflection and glare. Although these problems are not thought to produce chronic visual disorders, they can be sources of annoyance and possibly visual fatigue.

One recent study compared varying amounts of background surrounding luminance with subjective evaluations of asthenopia and specific objective measurements. The results of the study showed no significance in the value of surrounding luminance on the asthenopic symptoms for either CRT (cathode ray tube) or LCD (liquid crystal display) monitors. However, surrounding luminance was shown to significantly reduce the accommodation amplitude.⁹⁸ Another study revealed conflicting evidence concerning screen reflections. Glare was found to increase the amount of time required to read relatively easy passages but decreased the amount of time to read relatively difficult passages.²⁵

It was believed that because screen reflections are imaged behind the computer monitor, potentially conflicting cues could be created to initiate inappropriate accommodation responses and possibly affect blink rates.⁹⁰ Collins et al, however, found little evidence that reflections influence the accuracy of a user's accommodation response under binocular viewing conditions. They did, however, detect errors to a small degree (<0.25 D) under some monocular viewing conditions.¹⁴

In cases where it is not practical to reduce surrounding light, reduction of reflections and increase of contrast may be obtained from anti-glare filters.^{80,92} Whereas ambient light from the room passes

through the glare filter twice (once on the way in and once on reflection), direct light emitted from the VDT passes through the filter only once.⁸⁷ This increases the overall contrast of the picture as the background is attenuated more than the characters.

Recent studies demonstrate differing results in the efficacy of symptom relief with screen filters. One recent study revealed that of 60 full-time VDT workers, the 40 participants who used a screen filter reported less occurrence, shorter duration, and less intensive eye and musculoskeletal complaints after 1 month of use. The authors concluded that screen filters could improve the conditions for visual perception and thus relieve eyestrain.³⁸ Screen filters were also found to help schoolchildren with myopia that reported significant functional changes after 0.5 hours of computer usage, improving their overall functional indices.³² In contrast, a study of 25,064 participants investigated whether screen filters reduced the incidence of asthenopia with reference to weekly time spent at a VDT and duration of work at a VDT. The group showed that filters by themselves do not reduce the occurrence of asthenopia.⁷⁵

It is apparent that more research is needed to determine if screen filters are effective in the relief of ocular symptoms, and if so, which specific types of filters are most supportive in symptom relief.

C. REFRESH RATES

The refresh rate of a VDT represents the number of times per minute (measured in Hz) the screen is repainted to produce an image. If the refresh rate is too slow, the characters on the screen may appear to flicker. Flicker rate is of particular importance since the National Research Council reported that extremely low refresh rates (8 to 14 Hz) could induce epileptogenic seizures.⁶⁷ Perceived flicker has resulted in subjective complaints of annoyance, fatigue, and headache.¹⁶ The critical fusion frequency (CFF) is the refresh rate at which humans can no longer distinguish the pulsating beams of light as separate entities. In most viewing situations this rate is 30 to 50 Hz. The Video Electronic Standards Association (VESA) has recommended a minimum refresh rate of 75 Hz that minimizes flicker at all brightness levels. Berman et al supply evidence to perhaps support this recommendation with their study on human electroretinogram (ERG) responses. The authors clearly identified a synchronous ERG response for a VDT stimulus operating at 76 Hz.⁶

Studies have shown that much higher refresh rates may decrease ocular symptoms and increase user functionality. Jashchinski et al compared refresh rates of 300 Hz and the lowest frequency that did not produce visible flicker for each subject (50–90 Hz

in this study). At the lower refresh rate, mean accommodation in monocular vision was 0.06 D weaker, median eye blink duration was 6% shorter, and mean eye blink interval was 15% longer.⁴⁰ Another study determined that reading from a display at 500 Hz was 8 words per minute (3.05%) faster than at 60 Hz.⁵⁹ Kennedy and Murray revealed that lower refresh rates (50 Hz compared to 100 Hz) increase the number of prematurely triggered, less accurate saccades and an increase in the number of disrupted saccades in flight, which land short of their intended target.⁴⁵

Liquid crystal displays are quickly becoming the industry standard in monitor selection given their space conservation, bright screens, and extremely high refresh rates (essentially flicker-free). Ziefle recently published a user productivity study comparing CRT to LCD monitors. Search time (in milliseconds per line), fixation time (the time the eyes need to extract information), and fixation frequency per line (the number of fixations that are carried out to scan one line) were evaluated using both types of monitors. The LCD screens significantly provided the best performance in all three categories. She concluded that LCD is an important advance in screen technology, optimizing worker productivity and minimizing oculomotor effort and eyestrain in electronic reading (www.planar.com/reseller/pdfR/LCD_productivitywhite_paper.pdf).

D. RADIATION

The potential of health risk has persisted in the public eye concerning the claims that radiation emissions from VDTs could be responsible for hazardous effects to the computer user. Ionizing radiation is known to cause cellular changes and can affect living tissue through the breaking of chemical bonds and the charging of neutral molecules.⁷⁵ However, VDTs neither produce nor emit alpha, beta, gamma, or hard x-radiation. Small amounts of soft x-rays are produced, but almost all of this radiation is contained by the monitor's glass screen.¹⁶ Still, press reports continue to speculate that VDTs can potentially be responsible for skin problems, spontaneous abortions, and ocular disorders.

Numerous studies have shown that there is no evidence to support that VDT operators face health hazards or are exposed to electric, magnetic, or ionizing radiation fields significantly above ambient levels.^{20,61,82,87,94,103} Furthermore, Oftedal et al determined that a reduction in the electric field surrounding a VDT through the use of an electric-conducting screen filter did not reveal a significant reduction of eye symptom severity.⁶³ Kirsner and Federman published a recent review of the literature

and found that the data reviewed were either inconsistent or methodically flawed. They concluded that continued research should be performed to further define and elucidate the risk of electromagnetic radiation produced by VDTs.⁴⁸

V. Treatment

Without any doubt, the treatment of CVS requires a multidirectional approach due to the variety of complaints between users. When treating a patient, it is important to consider both ocular therapy as well as adjustment of the user's workstation and habits in an ergo-ophthalmologic approach.

A. LIGHTING

As mentioned earlier, proper lighting within the computer workstation area will enable the user to improve visual comfort and performance while eliminating annoyance and visual fatigue. An ideal environment would allow equalized brightness throughout the user's visual field (Abelson MB: *How to Fight Computer Vision Syndrome*. Rev Ophthalmol 114-6, 1999). Intense fluorescent lights can be diminished by removing a few of the lighting tubes. Excessive window lighting should be filtered with blinds, window coverings, or window tinting. If bright spots in the visual field cannot be avoided, shifting the workstation to a more favorable position may provide relief.⁷⁶

The actual type of lighting also appears to be important. One study focused on the visual work capacity with different sources of illumination. After comparing natural light, filament lamps, luminescent lamps, sodium lamps, and mercury-arc lamps, it was found that sodium lamps were the most adequate for high functional capacity of the visual analyzer.⁶²

Task lights use incandescent bulbs which are "warmer" (contain more red), are easier on the eyes, and cause less glare and eyestrain.¹² Task lights are often too bright, thus it is important to position the light carefully so that it does not throw bright light into the eyes or onto the computer screen producing glare.

As previously discussed, anti-glare filters may not reduce symptoms of asthenopia, but have been shown to reduce glare and improve contrast from the screen. This provides an effective means to eliminate reflections and therefore improve visual comfort.

B. VDT POSITIONING

Computer users often assume uncomfortable positions in order to properly view the screen. As previously mentioned, these postural distortions often

lead to pain in the back, neck, and shoulder. It is thus important to properly distance the monitor and maintain proper monitor height. Attempting to improve the physical ergonomics of the VDT workstation has been shown to reduce discomfort and improve performance.^{46,51,52}

Previously, it was recommended that the eye should be 16 to 30 inches from the screen.^{77,92} Distances outside of this range usually indicate a poor screen resolution or images that are too small. Recent data suggests that further distances may be more favorable to ocular symptoms. Three studies have compared visual strain with various screen distances at lengths of 66 cm (26.0 in.) versus 98 cm (38.6 in.); 50 cm (19.6 in.) versus 100 cm (39.4 in.); and 63 cm (24.8 in.) versus 92 cm (36.2 in.). In all three cases participants reported more eyestrain at the shorter distances from the monitor.⁴¹⁻⁴³ These studies suggest that distances of 35-40 inches may actually produce fewer complaints of visual strain.

It is also recommended that the screen should be placed 10 to 20 degrees below (or the middle of the screen 5-6 inches below) eye level. When the screen is higher than this, VDT users often tilt back their heads, causing muscle strain on the upper trapezius and neck muscles.^{80,92} Kietrys et al also report that a raised monitor has no beneficial effect of reducing postural stress of the cervical spine.⁴⁷ Lowering the monitor allows the VDT user to gaze downward, thus exposing less ocular surface to ambient air and reducing tear film loss.⁶⁵ Studies have shown that high screens result in greater eyestrain than low screens⁴² and that users actually prefer the low VDT position.³⁷

C. WORK BREAKS

Research has shown that when regular breaks are implemented, work efficiency actually improves, usually compensating for time lost on the break.^{24,28,56} The National Institute of Occupational Safety and Health found that short, frequent breaks demonstrated a decrease in worker discomfort and increase in productivity compared to the historical 15-minute morning and afternoon break.⁷⁶ Taking a quick walk around the office provides stretching of strained and fatigued muscles, a change of scenery, and possible relaxation.

Long periods of work without breaks are thought to be detrimental to ocular symptoms. In fact, one study showed that working for more than four hours at the VDT had a significant association with asthenopia.⁷² Frequent breaks are recommended to restore and relax the accommodative system, thereby preventing eyestrain. It is commonly believed that looking away at a distant object at least twice an hour

during computer usage is sufficient for prevention of visual fatigue.¹²

D. LUBRICATING DROPS

One of the most simple and therapeutic modes of therapy are lubricating eye drops intended to relieve the symptoms of dry eyes due to decreased blink rates. Abelson states, "An over-the-counter tear substitute can periodically rewet the ocular surface, contribute to tear volume, and maintain the proper balance of salts and acidity while viewing a terminal." It is important, though, to find the proper lubricating drop for the VDT user. A recent study in Japan revealed that the majority of self-medicating eye drop users were dissatisfied with the therapeutic effects.⁸¹ Another study indicates that higher viscosity eye drops may be more beneficial than balanced salt solutions. Although the higher viscosity drops did not vary blink rates, they normalized the interblink interval and relieved ocular discomfort more efficiently than balanced salt solutions following VDT use.¹ Unfortunately, these more viscous eye drops also cause a decrease in overall visual acuity.

E. COMPUTER EYEGLASSES

The occasional computer viewer may be able to get away with using their general eyewear, but those who spend more than one hour a day, including the occupational user, can benefit from the use of computer glasses. The trick is in choosing the right type.

Presbyopes have much to consider when deciding the right format of eyewear. Conventional bifocals are designed for viewing at 16 inches at an angle of 20 degrees or more below primary gaze. Computer screens are usually 24 inches away and only slightly below primary gaze.⁸⁰ General-wear progressive lenses are better, providing clear vision at an intermediate distance. Although one study has shown that users have preferred progressive lenses in the past,² the user must continually engage in a frustrating and fatiguing search for the perfect "sweet spot" on the lens that gives a clear view of the screen. This results in both annoyance and head/neck strain causing sore muscles. Occupational progressive lenses are now available which incorporate a large area in the top half of the lens for mid-distance viewing (i.e., VDT) and a bottom half of the lens for near distance (i.e., keyboard, desktop). Some lenses even contain a small area for distance viewing, usually at the top of the lens (Sheedy JE: Presbyopia and Computer Users. *Refract Eyecare Ophthalmol* 3:5-9, 1999). A recent study revealed that 24 symptomatic VDT users were significantly relieved from 7 of 10 symptoms reported (including neck/shoulder aches, eyestrain, and

blurred intermediates) following the use of occupational lenses.⁹

Eyeglasses can also provide a few other benefits in the treatment of CVS. Grant has suggested that the near triad of accommodation, convergence, and miosis should be expanded to include depression of gaze and extorsion.²⁹ The increase in extorsion observed on elevation of gaze at near point may induce binocular disruption and strain. Lazarus has shown that base-up and base-in prism could alleviate some of the CVS complaints because they decrease the elevation and convergence required. A double-blind study of 30 VDT users indicated a preference of prism and plus lenses than for plus lenses without prism.⁵⁰ Color-contrast optic filters are known to improve the color discriminating capacity when exposed to video display terminals. Feigin et al revealed that 20 of 23 subjects reported an improvement of visual fatigue after 4 weeks of using the eyeglasses with spectral filters.²³

We recently conducted a pilot study to analyze the ocular surface abnormalities in CVS in symptomatic subjects and compare them with non-symptomatic control population. A total of 88.2% of our symptomatic patients had meibomian gland dysfunction of >2 (meibomian gland dysfunction [MGD] was rated from 1 to 4 based on the quality of meibum expressed after applying pressure on the lids for 5 seconds) compared to 28.5% of controls (SD 1.26, $P = 0.001$). 58.8% of symptomatic subjects had meibomian gland dysfunction of >3. In the ongoing second phase of this study we are attempting to develop an effective treatment for CVS using microenvironment glasses (MEGS) which will specifically address the ocular surface pathology. The specialized MEGS we propose will create a controlled microenvironment for each eye, mounted on specially designed eyeglasses to increase the ambient humidity while blocking allergens and airborne irritants from the ocular surface. This increase in relative humidity around the eye will alter the tear film dynamics, by increasing the aqueous layer and decrease evaporation due to the increased humidity. Additionally, removing the accumulation of airborne particles and irritants will reduce frictional and foreign body irritation in the CVS patient. Lastly, the protected ocular surface prevents any irritation from direct drafts and air turbulence therefore improving the overall ocular surface integrity and comfort. The initial results are quite encouraging.

VI. Summary

The patient suffering from CVS can present to an eye care specialist in a variety of ways. A careful history and examination should reveal a correlation between

VDT usage and ocular complaints. It appears that the best treatment includes a multi-directional approach including modifications in the ergonomics of the workstation, eyeglasses correction, lighting and environmental factors, and properly scheduled work breaks from the video screen. To most of us, computers have become an irreplaceable necessity in our lives at work and home. As we continue to understand more about this syndrome, we will further protect our ocular health and thus prevent the proposed epidemic of the 21st century.

Method of Literature Search

PubMed was the primary source of abstract search with keywords of *computer vision syndrome, computers and eyestrain, computers and asthenopia, computers and ergonomics, video display terminals, computer monitors, computers and ergonomics, computers and liquid crystal display, and computers and dry eye*. Google was also used as a Web search engine to find any additional information of the above-mentioned keywords. All articles from the year 1970 to 2004 pertaining to summary of computer vision syndrome were included. Foreign abstracts and/or articles were accepted only if printed in English.

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